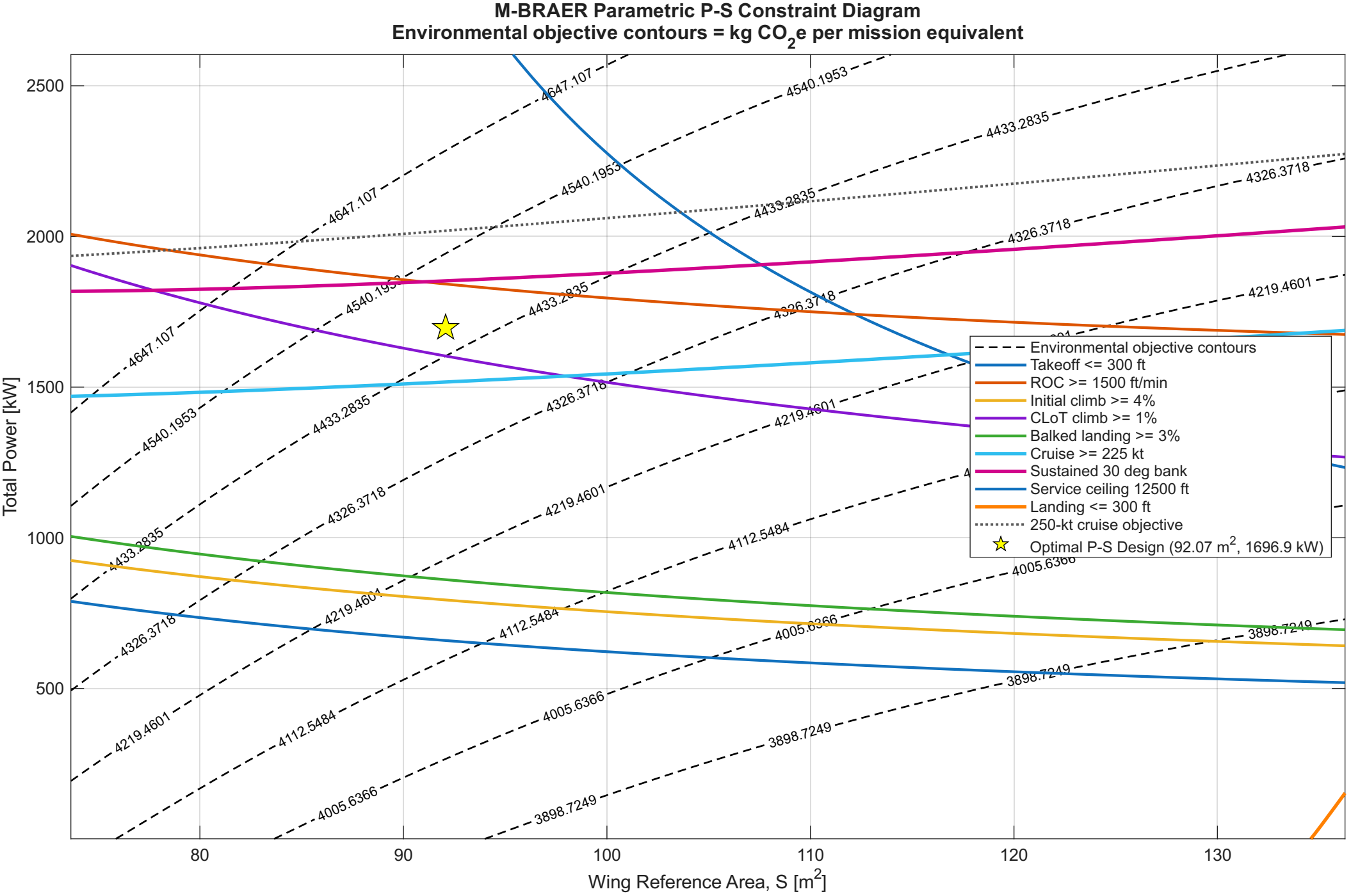
M-BRAER Parametric P-S Constraint Diagram
Environmental objective contours = kg CO2e per mission equivalent
Total Power [kW]
Wing Reference Area, S [m2]
2500
2000
1500
1000
500
80
90
100
110
120
130
4647.107
4540.1953
4433.2835
4326.3718
4219.4601
4112.5484
4005.6366
3898.7249
Environmental objective contours
Takeoff <= 300 ft
ROC >= 1500 ft/min
Initial climb >= 4%
CLoT climb >= 1%
Balked landing >= 3%
Cruise >= 225 kt
Sustained 30 deg bank
Service ceiling 12500 ft
Landing <= 300 ft
250-kt cruise objective
Optimal P-S Design (92.07 m2, 1696.9 kW)